

EXP.NO 1

Clustering based data analytics using R/Python. (K-Means, SOM algorithms)

AIM:

To perform clustering-based data analytics using K-Means and Self-Organizing Maps (SOM) to group similar data points into clusters

ALGORITHM:

K-means Algorithm

1. Choose number of clusters **K**
2. Initialize K centroids randomly
3. Assign each data point to nearest centroid
4. Recalculate centroids (mean of cluster points)
5. Repeat steps 3–4 until convergence

SOM Algorithm

1. Initialize SOM grid with random weights
2. Select an input vector
3. Find Best Matching Unit (BMU)
4. Update BMU and neighboring weights
5. Repeat for all data (multiple iterations)

SOURCE CODE:

```
# Import
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, adjusted_rand_score

# Load dataset
data = sns.load_dataset('tips')

# Features
X = data[['total_bill', 'tip', 'size']]

# Scaling
X_scaled = StandardScaler().fit_transform(X)
```

```

# ----- KMEANS -----
kmeans = KMeans(n_clusters=3, random_state=42)
k_labels = kmeans.fit_predict(X_scaled)

# Plot 1
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=k_labels)
plt.title("K-Means Clustering")
plt.xlabel("Total Bill")
plt.ylabel("Tip")
plt.show()

# ----- SOM -----
!pip install minisom -q
from minisom import MiniSom

som = MiniSom(10, 10, 3, sigma=1.0, learning_rate=0.5)
som.train_random(X_scaled, 500)

som_labels = [som.winner(x)[0]*10 + som.winner(x)[1] for x in X_scaled]

# Plot 2
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=som_labels)
plt.title("SOM Clustering")
plt.xlabel("Total Bill")
plt.ylabel("Tip")
plt.show()

# ----- EVALUATION -----
sil_k = silhouette_score(X_scaled, k_labels)
sil_s = silhouette_score(X_scaled, som_labels)

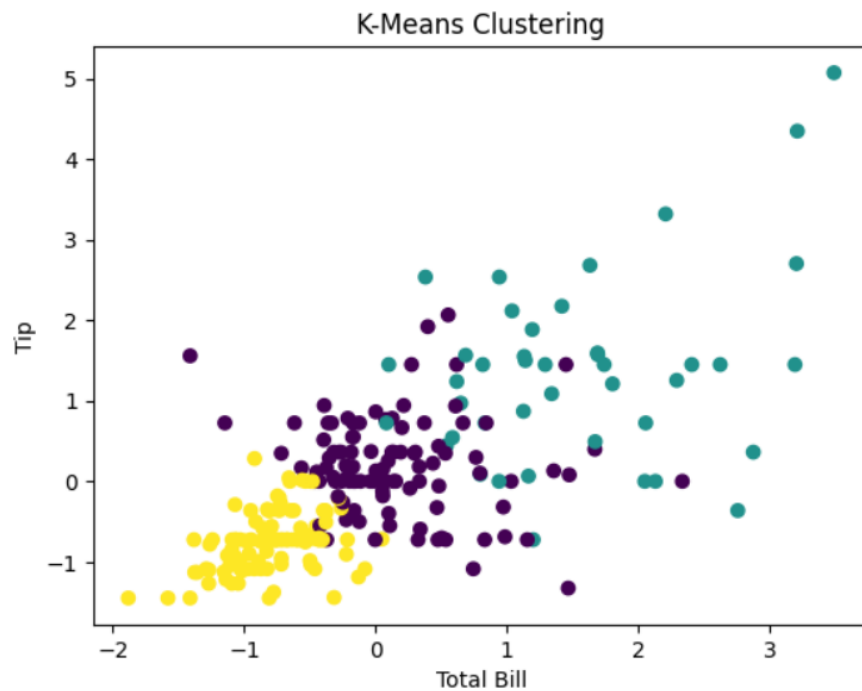
true_labels = pd.qcut(data['total_bill'], 3, labels=[0,1,2]).astype(int)

ari_k = adjusted_rand_score(true_labels, k_labels)
ari_s = adjusted_rand_score(true_labels, som_labels)

print("Silhouette KMeans:", sil_k)
print("Silhouette SOM:", sil_s)
print("ARI KMeans:", ari_k)
print("ARI SOM:", ari_s)

```

OUTPUT:



Preparing metadata (setup.py) ... done
Building wheel for minisom (setup.py) ... done



RESULT: Thus , the Clustering based data analytics using R/Python. (K-Means, SOM algorithms) successfully executed.